Note: If the application calls for a multipurpose grease with molybdenum and Cat Prime Application Grease is not available, consult the data sheets for the greases. Use a substitute that meets or exceeds the performance characteristics of Cat Prime Application Grease.

Severe Applications

The Caterpillar family of greases includes several which are made with a Calcium Sulfonate Complex thickener. This type of grease is ideal for severe applications. These greases provide more load carrying (galling resistance), lower wear, longer working life, exceptional water washout, and resistance to corrosion.

Hammer Grease

Note: Refer to the Operation and Maintenance Manual for a particular hammer for recommendations that relate to greases for Cat Hammers.

Cat Extreme Application Grease

Cat Extreme Application Greases are specially formulated in order to protect all of the most heavily loaded joints in any Cat machine against galling, wear, and corrosion. This protection is sustained while operating in moderate temperatures and with wet or dry working conditions.

Cat Extreme Application Greases are available in NLGI grades 1, and 2. Cat Extreme Application Greases are made with special blends of petroleum base oils and Calcium Sulfonate Complex thickener. These greases also have 5% Molybdenum diSulfide (MoS_2 or “Moly”) and performance enhancing additives.

A significant challenge exists in order to get grease to pump into the joints at low temperatures. Once the grease gets to the joint, the grease must have extremely high resistance to galling, wear, fretting, water washout, and corrosion in order to protect highly loaded joints adequately. Cat Extreme Application Greases are formulated with special blends of naphthenic petroleum base oils that have low pour points and will pump at lower temperatures. The ability to pump Cat Extreme Application Greases at lower temperatures means added insurance that all of the grease joints in the machine will be adequately lubricated even if the ambient temperature drops unexpectedly.

Even under severely loaded conditions, the grease should have a long working life. Inorder tomake greases that meet these greater demands, a Calcium Sulfonate Complex thickener with a properly blended naphthenic oil and/or a synthetic base oil is preferred. Caterpillar uses these ingredients in Cat Extreme Application Greases.

Cat Extreme Application Greases also have 5% Moly, instead of the 0% to 3% that is found in most other greases. This additional Moly greatly improves the ability of the grease to protect parts from damage in applications with severe impact (slamming).

Cat Extreme Application Greases are also made to be extra tacky. In some applications, the film of grease must adhere to vertical surfaces. An example of this application is swing gears for excavators. Many conventional greases do not have enough of the tacky characteristic to keep the grease in place resulting in incomplete lubrication and premature wear or failure of components.

Note: If the application calls for Cat Extreme Application Grease and Cat Extreme Application Grease is not available, consult the data sheets for the grease. From these data sheets, use a substitute that meets the performance characteristics of Cat Extreme Application Grease.

Cat Extreme Application Grease-Desert

Cat Extreme Application Grease-Desert is formulated to protect the most severely loaded joints in Cat machines against galling, wear, and corrosion. This protection is sustained while working in moderate to hot temperatures with wet or dry conditions.

Cat Extreme Application Grease-Desert is an NLGI grade 2 grease. This grease is made with a high viscosity petroleum base and a Calcium Sulfonate Complex thickener. This grease also has 5% Molybdenum diSulfide (MoS_2 or “Moly”) and tackifier.

As the temperature changes, Cat Extreme Application Grease-Desert experiences minimal change in viscosity and will maintain a thick lubricant film even at hot temperatures.

Cat Extreme Application Grease-Desert is made with a Calcium Sulfonate Complex thickener. Cat Extreme Application Grease-Desert provides the necessary protection against galling, wear, fretting, water washout, and corrosion. Cat Extreme Application Grease-Desert also has a long life. This grease will resist breakdown even when the application is under extremely heavy loads and with frequent oscillations. This protection is sustained while work is being done at moderate to hot temperatures under wet or dry conditions.

Cat Extreme Application Grease-Desert has 5% molybdenum instead of the 0% to 3% that is found in most other greases. This additional molybdenum greatly improves the ability of the grease to protect parts from damage in applications with severe impact (slamming).

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Cat Extreme Application Grease-Desert is made to be extra tacky. In some applications, the film of grease must adhere to vertical surfaces. An example of this application is swing gears for excavators. Many conventional greases do not have enough of the tackifier to allow the grease to stay in place resulting in incomplete lubrication and premature wear or failure of components. In addition, many of these conventional greases do not have the performance, particularly at high ambient temperatures, to adequately protect the gear teeth in these applications.

Cat Extreme Application Grease-Desert prevents galling and wear in hot temperatures under extremely severe loads and conditions. In moderate temperatures, Cat Extreme Application Grease-Desert can be used in extremely severe applications if an improvement over Cat Extreme Application Grease is desired. Cat Extreme Application Grease-Desert can be used in applications that require the lubricant to last for long periods of time because this grease has an extremely high performance and long life.

Note: If the application calls for Cat Extreme Application Grease-Desert and Cat Extreme Application Grease-Desert is not available, consult the data sheets for the grease. From these data sheets, use a substitute that meets the performance characteristics of Cat Extreme Application Grease-Desert.

Cat Extreme Application Grease-Arctic (Semi-Synthetic)

Cat Extreme Application Grease-Arctic is formulated to protect the most heavily loaded joints in Caterpillar machines against galling, wear, and corrosion. This grease resists breakdown even under extremely heavy loads and/or in applications with frequent oscillations. This grease provides protection that will be sustained for long periods of time and in wet or dry conditions. This protection is sustained, while work is being done in moderate temperatures and in low temperatures that may reach −50° C (−58° F). In moderate temperatures, Cat Extreme Application Grease-Arctic can be used in the application if the compartment is sealed tightly to contain the grease

Cat Extreme Application Grease-Arctic is available in NLGI grade 0. This grease is made with a semi-synthetic base oil that has a low viscosity for optimal cold-weather performance. The performance is enhanced with 5% Molybdenum diSulfide (MoS_2 or “Moly”) and tackifier.

Cat Extreme Application Grease has low viscosity and semi-synthetic base oil. Hence, as the temperatures drop, this grease has minimal change of viscosity and continues to flow easily. Cat Extreme Application Grease-Arctic pumps easily at extremely low temperatures. In fact, Cat Extreme Application Grease-Arctic NLGI grade 0.5 can be pumped through standard automatic lubrication systems that are machine mounted and at temperatures down to −50° C (−58° F). This means that the grease can be pumped through long unheated lines and into the required joints.

Cat Extreme Application Grease-Arctic has 5% of molybdenum instead of 0% or 3% that is found in most of the other greases. This additional molybdenum greatly improves the ability of the grease in order to protect parts from damage in applications with severe impact (slamming).

Cat Extreme Application Grease-Arctic is made to be extra tacky. In some applications, the film of grease must adhere to the vertical surfaces. An example of this application is swing gears for excavators. Many conventional greases do not have enough tackifier to adhere well to the vertical surfaces. This ability is necessary in order to protect the gear teeth in these applications adequately.

Cat Extreme Application Grease-Arctic may be used in applications that require the lubricant to last for long periods of time. This grease has an extremely high performance and also has a long life.

If the application calls for Cat Extreme Application Grease-Arctic and no Cat Extreme Application Grease-Arctic is available, consult the data sheets for the grease. Use a substitute that most closely meets the performance characteristics.

Cat Ball Bearing Grease

This grease is recommended for applications that utilize roller bearings and ball bearings at low to moderate loads and at high speed. Typical applications for this grease are electric motors, alternators, and constant velocity (CV) joints for automotive products. Cat Ball Bearing Grease is an NLGI grade 2 grease. This grease is made with petroleum base oil and polyurea thickener.

Grease Application Charts

Types of Grease

Severe Applications are heavy loads, frequent oscillations, and heavy shock loads.

Extreme pressure (EP) refers to 4-Ball Weld point in the technical data sheet (“ASTM D 2596”).

N/R = Not Recommended

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Table 48

Cat Grease Recommendations by Application						
Cat Grease Name	**NLGI Grade**	**Severe Applications**	**Severe Applications**	**Severe Applications**	**Severe Applications with Extremely Heavy Loads**	**Severe Applications**
Operating Conditions		Extremely Low to Low Temps	Low to Moderate Temps	Moderate to High Temps	Moderate to High Temps	High to Extremely High Temps
Optimum Temperature Range		−50° C (−58° F) to −18° C (0° F)	−23° C (−10° F) to 29° C (85° F)	18° C (65° F) to 60° C (140° F)	18° C (65° F) to 41° C (105° F)	38° C (100° F) to 232° C (450° F)
Cat Extreme Application Grease-Desert	2	N/R	Excellent	Excellent	Excellent	Very Good
Cat Extreme Application Grease	2	N/R	Excellent	Excellent	Very Good	Good
Cat Extreme Application Grease	1	Good	Excellent	Good	Good	Fair
Cat Extreme Application Grease-Arctic	0	Excellent	Very Good	N/R	N/R	N/R
Cat Prime Application Grease	2	N/R	Fair	Good	Good	N/R
Cat Utility Grease	2	N/R	N/R	N/R	N/R	N/R
Cat Ball Bearing Grease	2	N/R	N/R	N/R	N/R	N/R
Hammer Paste	2	N/R	Good	Excellent	Excellent	Excellent

Pumpability is based on “USS Mobility and Lincoln Ventmeter Tests”. Performance may vary depending on lubrication equipment and the length of the lines.

Service Life refers to the percent of change after 100,000 strokes in technical data sheet (“ASTM D 217”).

Corrosion protection was measured per “ASTM D1743 Rust preventative Properties” and “ASTM D4048 Copper Corrosion Test”.

Water washout resistance refers to roll stability with water and the percent of change (“ASTM D 1264”).

For additional performance data on these greases refer to this Special Publication, “Reference Material” article.

Table 49

Type of Cat Grease						
Cat Grease Name	**NLGI Grade**	**Low Temp. Pumpability**	**Service Life**	**Corrosion Protection**	**Water Washout Resistance**	**Extreme Pressure (EP)**
Cat Extreme Application Grease-Desert	2	above 2° C (35° F)	Excellent	Excellent	Excellent	Excellent
Cat Extreme Application Grease	2	above −7° C (20° F)	Excellent	Excellent	Excellent	Excellent

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(Table 49, contd)

Type of Cat Grease						
Cat Grease Name	**NLGI Grade**	**Low Temp. Pumpability**	**Service Life**	**Corrosion Protection**	**Water Washout Resistance**	**Extreme Pressure (EP)**
Cat Extreme Application Grease	1	above −18° C (0° F)	Excellent	Excellent	Excellent	Excellent
Cat Extreme Application Grease-Arctic	0	above −43° C (−45° F)	Excellent	Excellent	Excellent	Excellent
Cat Prime Application Grease	2	above −18° C (0° F)	Good	Fair	Fair	Very Good
Cat Utility Grease	2	above −23° C (−10° F)	Good	Fair	Fair	Fair
Cat Ball Bearing Grease	2	above −18° C (0° F)	Very Good	Good	Good	N/R
Hammer Paste	2	N/R	Good	Fair	Fair	Excellent

Type of Cat Grease by Machine Category

Drive shaft universal joints should NOT be serviced with any lubricant that contains Molybdenum disulfide (Extreme Application or Prime Application).

Table 50

Type of Cat Grease by Machine Category									
Vehicle	**Application Point**	**Typical Load and Speed**	**Load Factor**	**Ambient Temperature Range**				**NLGI Grade**	**Grease Type**
				° C		**° F**			
				Min	**Max**	**Min**	**Max**		
Agricultural Products	Front Axle, Drive Shaft Slip Spline	High	Agricultural drawbar work at full throttle, engine lugged to max. power most of the time, little or no idling or travel in reverse.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Agricultural drawbar work at full throttle but not always lugging engine. Some idling and some travel with no load.	−20	40	−4	104	2	Prime Application Grease
		Low	Considerable idling or travel with no load.	−30	40	−22	104	2	Utility Grease
Asphalt Pavers	Track Takeup Linkage, Takeup Idler Bearings	High	Wide width, deep lift paving.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	3-4 m (10'-12') width, 50-75 mm (2-3) lift.	−20	40	−4	104	2	Prime Application Grease
		Low	Narrow width paving - low production.	−30	40	−22	104	2	Utility Grease

(continued)

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(Table 50, contd)

Type of Cat Grease by Machine Category									
Vehicle	**Application Point**	**Typical Load and Speed**	**Load Factor**	**Ambient Temperature Range**				**NLGI Grade**	**Grease Type**
				° C		**° F**			
				Min	**Max**	**Min**	**Max**		
Backhoe Loaders	Drive Shaft Slip Spline, Kingpin Bearing, Wheel Bearing	High	Production work with long cycles and/or constant flow implements.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	General work with regular cycles in medium applications.	−20	40	−4	104	2	Prime Application Grease
		Low	Utility work with intermittent cycles in light to medium applications.	−30	40	−22	104	2	Utility Grease
Vibratory Soil and Asphalt Compactors	Articulation Bearing, Control Cables, Drum Bearings, Leveling Blade, Operator Platform, Oscillating Bearings, Oscillating Pins, Steering Cylinder Ends, Steering Pins, Weight Bearings	High	Vibration 80-100%, heavy cohesive soil, 350 mm (12) lifts or more.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Vibration 50-80%, granular soil, 100mm-305 mm (4-12) lifts.	−20	40	−4	104	2	Prime Application Grease
		Low	Vibration 30-50%, asphalt mix, 51mm - 100 mm (2-4) lifts.	−30	40	−22	104	2	Utility Grease
Wheeled and Track-type Excavators	Axle Bearings, Belt Tensioner, Blade Linkage, Boom, and Stick Linkage, Boom Cylinder Bearing, Boom Cylinder Head, Cab Riser, Drive Shaft Slip Spline, Drive Shaft Support Bearing, Fan Drive Bearing, Steering Linkage, Stick Linkage, Swing Bearings, Swing Drive Gear, Swing Frame	High	Most pipeline applications in hard rocky material. Digging 90-95% of the daily work schedule.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Most residential sewer applications in natural bed clay. Digging 60-85% of the daily work schedule. Most log loading applications.	−20	40	−4	104	2	Prime Application Grease
		Low	Most utility, urban applications in sandy loam. Digging less than 50% of the daily work schedule. Scrap handling applications.	−30	40	−22	104	2	Utility Grease

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(Table 50, contd)

Type of Cat Grease by Machine Category									
Vehicle	Application Point	Typical Load and Speed	Load Factor	Ambient Temperature Range				NLGI Grade	Grease Type
				° C		° F			
				Min	Max	Min	Max		
Forest Products	Articulation Bearings, Bogie Bearings, Drive Shaft Support Bearing, Drive Shaft Slip Spline, Gate Cylinder, Grapple Head Rotator, Swing Drive Bearing, Swing Drive Gear, Winch Drive Shaft Bearing, Winch Drive Shaft Support Bearing, Winch Drum Bearing	High	517- Skidding over 4536 kg (10,000 lb) in steep terrain (over 30%) with high resistance. 525B - Skidding over 6800 kg (15,000 lb) in steep terrain (over 10%) with high resistance. 527 - Skidding over 6360 kg (14,000 lb) in steep terrain (over 30%) with high resistance.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	517- Skidding up to 4536 kg (10,000 lb) in moderate terrain (8-30%) with medium resistance. 525B - Skidding up to 6800 kg (15,000 lb) in moderate terrain (5-10%) with average resistance. 527 - Skidding up to 6360 kg (14,000 lb) in moderate terrain (8-30%) with medium resistance.	−20	40	−4	104	2	Prime Application Grease
		Low	517 - Skidding less than 4536 kg (10,000 lb) in flat terrain (0-8%) with low resistance. 525B - Skidding less than 4500 kg (10,000 lb) in flat terrain (0-5%) with low resistance. 527 - Skidding less than 6360 kg (14,000 lb) in flat terrain (0-8%) with low resistance.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease
Motor Graders	Articulation Bearings, Articulation Pins, Blade Lift Cylinder Socket, Centershift Cylinder Socket, Fan Drive Belt Tightener, Pump Drive Shaft Slip Spline	High	Ditching, fill spreading, spreading base material, ripping, heavy road maintenance, snow plowing.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Average road maintenance, road mix work, scarifying, snow plowing.	−20	40	−4	104	2	Prime Application Grease

(continued)

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(Table 50, contd)

Type of Cat Grease by Machine Category									
Vehicle	Application Point	Typical Load and Speed	Load Factor	Ambient Temperature Range				NLGI Grade	Grease Type
				° C		° F			
				Min	Max	Min	Max		
		Low	Finish grading, light maintenance, road travel.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease
Off-Highway Trucks and Tractors	Belt Tightener, Drive Shaft Slip Spline, Fan Drive Pulley, Hoist Control Bellcrank, Hoist Cylinder Bearing, Front and Rear Axle A-Frame Bearing, Steering Cylinder End, Steering Linkage, Steering Tie Rod and Pin Bearings, Swaybar, Tachometer Drive	High	40-50% load factor.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	30-40% load factor.	−20	40	−4	104	2	Prime Application Grease
		Low	20-30% load factor.	−30	40	−22	104	2	Utility Grease
	Autolube System			−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
	Fan Drive Bearings Traction Motor Bearings			−20	40	−4	104	2	Ball Bearing Grease
Articulated Trucks	Belt Tightener, Drive Shaft Slip Spline, Fan Drive Pulley, Hoist Control Bellcrank, Hoist Cylinder Bearing, Rear Axle A-Frame Bearing, Steering Cylinder End, Steering Linkage, Steering Tie Rod and Pin Bearings, Tachometer Drive, Tailgate Bearings	High	Long haul time with frequent adverse grades. Continuous use on poorly maintained haul roads with high rolling resistance.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Normal load and haul time. Varying load and haul road conditions. Some adverse grades. Some high rolling resistance	−20	40	−4	104	2	Prime Application Grease
		Low	Large amount of idling. Short to medium hauls on well maintained level haul roads. Minimum total resistance.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease

(continued)

(Table 50, contd)

Type of Cat Grease by Machine Category									
Vehicle	Application Point	Typical Load and Speed	Load Factor	Ambient Temperature Range				NLGI Grade	Grease Type
				° C		° F			
				Min	Max	Min	Max		
Pavement Profilers/Road Reclaimers	Chain Tensioner, Fan Drive Bearings, Propel Transmission Control Cable, Pump Drive Shaft Slip Spline, Rotor Transmission Shift Linkage, Wheel Hub Bearings	High	457 mm (18) soil/305 mm (12) asphalt.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	305 mm (12) soil/150 mm (6) asphalt.	−20	40	−4	104	2	Prime Application Grease
		Low	150 mm (6) soil/100 mm (4) asphalt.	−30	40	−22	104	2	Utility Grease
Telehandlers	Boom Chain, Boom Cylinder Pin, Boom Extension and Retraction Chain Pulley, Boom Head Section, Boom Pivot Shaft, Compensating Cylinder Bearing, Drive Shaft Slip Spline, Fork Leveling Cylinder Pin, Frame Leveling Cylinder Pin, Stabilizer, and Cylinder Bearings	High	Heavy loading, continuous loading with steady cycling.	−35	40	−31	104	1	Ultra 5Moly Grease
				−30	50	−22	122	2	
		Medium	Normal load and work conditions.	−20	40	−4	104	2	Advanced 3Moly Grease
		Low	Considerable idling or travel with no load.	−30	40	−22	104	2	Multipurpose Grease
	Boom Wear Pad			−15	48	5	118	0	Ultra 5Moly Grease
	Grease Fittings			—15	50	5	122	2	Ultra 5Moly Grease
Track Loaders	Equalizer Bar Pin, Recoil Piston, Track Adjustment	High	Continuous excavating and loading from bank. Land clearing.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Bank or stockpile loading with idling periods. Load and carry.	−20	40	−4	104	2	Prime Application Grease
		Low	Large amounts of idling in any application.	−30	40	−22	104	2	Utility Grease
Track-Type Tractors	Adjust Track, Angle Blade Tilt Brace, Equalizer Bar Pins, Fan Drive Belt Tightener, Lift Cylinder Yoke Bearing, Winch Drum Bearing, Winch Fairlead Rollers	High	Ditching, fill spreading, spreading base material, ripping, heavy road maintenance, snow plowing.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Average road maintenance, road mix work, scarifying, snow plowing.	−20	40	−4	104	2	Prime Application Grease
		Low	Finish grading, light maintenance, road travel.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease

(continued)

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(Table 50, contd)

Type of Cat Grease by Machine Category									
Vehicle	Application Point	Typical Load and Speed	Load Factor	Ambient Temperature Range				NLGI Grade	Grease Type
				° C		° F			
				Min	Max	Min	Max		
Wheel Dozers and Compactors	Articulation Bearings, Drive Shaft Support Bearing, Drive Shaft Slip Spline, Lift Cylinder Yoke Bearing, Steering Shaft, Tilt Cylinder Bearing	High	Heavy dozing, compacting heavy material. Heavy landfill work.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Production dozing, most push-loading, shovel cleanup, normal compaction.	−20	40	−4	104	2	Prime Application Grease
		Low	Considerable idling or travel with no load.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease
Wheel Loaders	Articulation Bearings, Axle Oscillation Bearings, Bucket Pivot Bearings, Drive Shaft Support Bearing, Drive Shaft Slip Spline, Frame Pivot Bearings, Steering Cylinder Bearings, Steering Shaft, Wheel Brake Camshafts	High	Continuous and aggressive tight truck loading, hard bank excavation, and shot rock loading from a face.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Maximum productivity in aggregate truck loading and hopper charging. Assumes the normal slamming and load and carry associated with the high productivity stockpile load-out and batch plant applications.	−20	40	−4	104	2	Prime Application Grease
		Low	Light utility, construction, low production aggregate truck loading, and most logging applications where there is considerable idling, empty traveling, and load and carry.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease
Integrated Toolcarriers	Articulation Bearings, Axle Oscillation Bearings, Bucket Pivot Bearings, Drive Shaft Support Bearing, Drive Shaft Slip Spline, Frame Pivot Bearings, Steering Cylinder Bearings, Steering Shaft, Wheel Brake Camshafts	High	Continuous and aggressive tight truck loading, hard bank excavation, and shot rock loading from a face.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	

(continued)

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(Table 50, contd)

Type of Cat Grease by Machine Category									
Vehicle	**Application Point**	**Typical Load and Speed**	**Load Factor**	**Ambient Temperature Range**				**NLGI Grade**	**Grease Type**
				° C		**° F**			
				Min	**Max**	**Min**	**Max**		
		Medium	Maximum productivity in aggregate truck loading and hopper charging. Assumes the normal slamming and load and carry associated with the high productivity stockpile load-out and batch plant applications.	−20	40	−4	104	2	Prime Application Grease
		Low	Light utility, construction, low production aggregate truck loading, and most logging applications where there is considerable idling, empty traveling, and load and carry.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease
Wheel Tractor-Scrapers	Brake Camshaft Bearings, Drive Shaft Slip Spline, Ejector Carrier, and Support Rollers, Sprocket Shaft Support Bearings	High	Continuous high total resistance conditions with steady cycling.	−35	40	−31	104	1	Extreme Application Grease
				−30	50	−22	122	2	
		Medium	Typical road building use.	−20	40	−4	104	2	Prime Application Grease
		Low	Average use but with considerable idling, favorable grades, low rolling resistance, and easy loading material.	−30	40	−22	104	2	Utility Grease
	Fan Drive Bearings			−20	40	−4	104	2	Ball Bearing Grease
Hammers	Hammer Lubrication							2	Hammer Paste
Extreme Temperature Conditions - All Applications Except Hammers	All except Autolubrication Systems, Fan Drive Bearings, and Universal Joints	High, Medium, and Low		−50	20	−58	68	0.5	Extreme Application Grease-Arctic
				−20	60	−4	140	2	Extreme Application Grease-Desert
Hydro-Mechanical Work Tools		High		-35	40	-31	104	1	Extreme Application Grease
				-30	50	-22	122	2	

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Fuel Specifications

i08021832

General Fuel Information

SMCS Code: 1250; 1280

NOTICE
Every attempt is made to provide accurate, up-to-date information. By the use of this document, you agree that Caterpillar Inc. is not responsible for errors or omissions.

The information provided are the latest recommendations for the Caterpillar diesel engines that are covered by this Special Publication. This information supersedes all previous recommendations which have been published for the Caterpillar diesel engines that are covered by this Special Publication. Special fluids are required for some engines and continued use of these special products will be necessary. Refer to the applicable Operation and Maintenance Manual.

This publication is a supplement to the Operation and Maintenance Manual. This publication does not replace the engine-specific Operation and Maintenance Manuals.

NOTICE
These recommendations are subject to change without notice. Consult your local Cat dealer for the most up to date recommendations.

NOTICE
In order to avoid potential damage to your Cat machine and/or Cat engine, only purchase Cat fluids and Cat filters through your Cat dealer or Cat authorized outlets. For a list of authorized Cat parts outlets in your area, consult your Cat dealer.

If you purchase what appear to be Cat fluids and/or Cat filters through other outlets/sources, you are at a very high risk of purchasing counterfeit ("look-alike") products.

Counterfeit or "look-alike" products may visually appear the same as the original Cat product, but the product performance and internal quality will typically be very low.

Counterfeit or "look-alike" products have a very high likelihood of causing and/or allowing engine and/or machine compartment damage.

NOTICE
Many of the guidelines, recommendations, and requirements that are provided in this Special Publication are interrelated. Before using the provided information, it is the responsibility of the user of this Special Publication to read and understand the information provided in its entirety.

It is the responsibility of the user of this Special Publication to follow all safety guidelines found in this Special Publication and in engine and/or machine specific Operation and Maintenance Manual when performing all recommended and/or required engine, engine systems, and/or machine maintenance.

For questions concerning the information presented in this Special Publication and/or in your product Operation and Maintenance Manual, and/or for additional guidelines and recommendations (including maintenance interval recommendations/requirements) consult your Cat dealer.

Follow all industry standard safety practices when operating engines and/or machines and when performing all recommended and/or required maintenance.

NOTICE
Commercial products that make generic claims of meeting "Cat" requirements without listing the specific Cat recommendations and requirements that are met may not provide acceptable performance. Commercial products may cause reduced engine and/or machine fluid compartment life. Refer to this Special Publication and refer to product specific Operation and Maintenance Manual for Cat fluids recommendations and requirements.

Note: Instructions for the installation of the filter are printed on the side of each Caterpillar spin-on filter. For nonCaterpillar filters, refer to the installation instructions that are provided by the supplier of the filter.

PROOF

NOTICE
In order to meet expected fuel system component life, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with common-rail fuel systems. Also, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with unit injected fuel systems. For all other Cat diesel engines (mostly older engines with pump, line and nozzle type fuel systems), the use of 4 micron(c) absolute or less secondary fuel filtration is strongly recommended. Note that all current Cat diesel engines are factory equipped with Cat Advanced Efficiency 4 micron(c) absolute fuel filters.

In order to obtain additional information on Cat designed and produced filtration products, refer to the "Reference Material" article, "Filters" and "Miscellaneous" topics in this Special Publication. Consult your Cat dealer for assistance with filtration recommendations for your Cat machine.

NOTICE
Caterpillar does not warrant the quality or performance of non-Caterpillar fluids and filters.

Diesel Fuel and Injector Health

Fuel injectors are highly engineered components built with very tight clearances and designed to spray precise amounts of fuel into the combustion chamber. The accurate operation of the fuel injectors supports the performance and noise of the engine and impacts the emissions.

Fuels that are not per the characteristics described in this chapter and in the recommended specifications can lead to deposits in the injector, cold start issues, smoke, noise, low performance, increased emissions, and other issues.

Deposits in the injector can be external or internal:

- External deposits form on the tip of the injector. These deposits become carbonaceous (mainly composed of carbon) due to the high temperature of the combustion chamber. The deposits prevent the appropriate fine spray of the fuel and the proper atomization in the combustion chamber. These issues can significantly degrade the operation of the engine.

- Internal deposits form on the internal moving components of the injector. These deposits close the tight clearances in the injector and prevent the components from moving as designed. Some internal deposits can be "soaps" that form due to reaction of sodium contaminants in the fuel with certain fuel additives. Sometimes, these soaps cause injector sticking in the open or closed positions. The result is a malfunction of the injector.

Cleanliness of the injectors impact the performance, power, fuel consumption, and emissions throughout the life of modern diesel engines. To keep the injectors clean, avoid injector deposits and ensure appropriate operation and long life of the injectors:

- Ensure that the fuel has the proper chemistry. Fuels that are per the specifications given in this chapter have the appropriate cetane value, viscosity, density, distillation, stability, lubrication, and energy content.

- Ensure that the fuel has the proper quality. Fuel quality is determined by the lack of contamination and water. This quality is ensured through following the contamination control recommendations and filtering the fuel as recommended by the Operation and Maintenance Manual and as stated in this chapter.

- The use of Fuel Additives to improve detergency and overall quality of fuels may be needed sometimes. When needed, Caterpillar recommends the use of Cat Fuel Conditioner and Cat Fuel System Cleaner. Caterpillar cannot validate or recommend other additives available in the market. The fuel supplier should be consulted when using additives to enhance other properties of the fuel.

Consult with the engine or machine Operation and Maintenance Manual for any special fuel requirements.

Consult with your fuel supplier to ensure that the fuel follows all the recommendations given in this chapter.

General Recommendations and Contamination Control Guidelines for Fuels

Follow all applicable industry standards and all applicable governmental, environmental, and safety guidelines, practices, regulations, and mandates.

Note: These general recommendations and guidelines concerning maintenance and care of fuel and fuel storage systems are not intended to be all inclusive. Discuss proper fuel safety and health, handling, and maintenance practices with your fuel supplier. Use of these general recommendations and guidelines does not lessen the engine owners and/or fuel supplier responsibility to follow all industry standard practices for fuel storage and for fuel handling.

Note: Where recommendations for draining water and/or sediment and/or debris are stated, dispose of this waste according to all applicable regulations and mandates.

PROOF

Note: Caterpillar filters are designed and built to provide optimal performance and protection of the fuel system components.

Clean fuels, as detailed below, are strongly recommended to allow optimal performance and durability of the fuel systems and to reduce power loss, failures, and related down time of engines.

Fuels of “ISO 18/16/13” cleanliness levels or cleaner as dispensed into the engine or machine fuel tank should be used. Reduced power, failures and related downtime can result if clean fuels are not used. Fuels of “ISO 18/16/13” are particularly important for new fuel system designs such as Common Rail injection systems and unit injection systems. These new injection system designs utilize higher fuel pressures and are designed with tight clearances between moving parts to meet required stringent emissions regulations. Peak injection pressures in current fuel injection systems may exceed 30,000 psi. Clearances in these systems are less than 5 µm. As a result, particle contaminants as small as 4 µm can cause scoring and scratching of internal pump and injector surfaces and of injector nozzles.

Water in the fuel causes cavitation, corrosion of fuel system parts, and provides an environment where microbial growth in the fuel can flourish. Other sources of fuel contamination are soaps, gels, or other compounds that may result from undesirable chemical interactions in the fuels. Gels and other insoluble compounds can also form in biodiesel fuel at low temperatures or if biodiesel is stored for extended periods. An indication of microbial contamination, detrimental fuel additives interactions, or cold temperature gel is very rapid filter plugging of bulk fuel filters or machine fuel filters.

To reduce downtime due to contamination, follow these fuel maintenance guidelines in addition to the recommendations given in the “Contamination Control” Chapter in this Special Publication:

- Use high-quality fuels per recommended and required specifications (refer to the “Fuel” chapter in this Special Publication).
- Do not add new engine oil, waste engine oil or any oil product to the fuel unless the engine is designed and certified to burn diesel engine oil (for example Caterpillar ORS designed for large engines). Engine oils may raise the sulfur level of the fuel and may cause fouling of the fuel system and loss of performance. Engine oils in fuels can also reduce the maintenance intervals of aftertreatment devices in Tier 4 machines.
- Use recommended Cat filtration products, including Cat Advanced Efficiency Fuel Filters. Change your fuel filters per recommended service requirements or as needed. Never fill the new secondary and tertiary fuel filter with fuel before installation. Use the fuel priming pump to remove air from the system.
- Follow proper practices of fuel transport and filtration from storage tank to the machine to allow the delivery of clean fuel to machine tank. Keep the fuel storage tank clean of water, debris, and sediment.
- Filter the fuel coming into the bulk storage fuel tank and at every subsequent transfer into and out of any container and prior to adding to the engine fuel tank preferably through filters with a rating of 20 microns (c) absolute or less. The use of wire mesh media (strainer-type filters) is NOT recommended except when filters with standard media (cellulose or synthetic) are downstream of the wire mesh media filters. Wire mesh filters typically have poor filtration efficiency and can corrode with time, allowing the passing of large particles.
- Caterpillar recommends the use of properly designed and grounded bulk fuel filter / coalescer units which remove both particulate contamination and water in a single pass. These units can clean fuel to “ISO 16/13/ 11” or cleaner and can remove free water to 200 ppm (mg/kg) or less.
- Cat offers heavy duty filter / coalescer units to accommodate fueling rates from 50 to 300 gpm (gallons per minute). Cat custom designs filter / coalescer units specifically for the conditions of fuel at the worksite if needed. Refer to Special Publication, PEHJ0156, “Cat Bulk Fuel Filtration Systems”, and consult your Cat dealer for availability of bulk filtration products.
- Fill machine fuel tanks with fuels of “ISO 18/16/13” cleanliness level or cleaner, in particular for engines with common rail and unit injection systems. When you refuel the machine, filter the fuel through a 4 µm absolute filter (Beta = 75 up to 200 at 4 microns) to reach the recommended cleanliness level. This filtration should be located at the device that dispenses the fuel to the engine or machine fuel tank. In addition, filtration at the dispensing point should have the ability to remove water to ensure that fuel is dispensed at 200 ppm water or less.
- Keep the area around the fuel tank filler neck clean of debris to prevent dirt entry and contamination of the fuel tank.
- Drain your water separators daily per the Operation and Maintenance Manual of your machine.
- Install desiccant type breathers of 4 µm or less absolute efficiency with the ability to remove water on bulk storage tanks.

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- Drain your fuel tanks of sediment every 500 hours or 3 months per the Operation and Maintenance Manual of your machine.
- Centrifugal filters may need to be used as a prefilter with fuel that is severely contaminated with gross amounts of water or large particulate contaminants. Centrifugal filters can effectively remove large contaminants but may not be able to remove the very small abrasive particles required to achieve the recommended “ISO” cleanliness level. Bulk filter / coalescers are necessary as a final filter to achieve the recommended cleanliness level.
- Cover, protect, and ensure cleanliness of all connection hoses, fittings, and dispensing nozzles.
- Test for microbial contamination regularly and take proper corrective action if contamination is present. Properly dispose of cleanup waste according to all applicable local regulations and mandates.
- Every 3 months, or sooner if problems are suspected, analyze the fuel for acid number, density, particle content, water and microbial growth (tank bottom fuel sample). More tests can also be run per the “Caterpillar Specification for Distillate Diesel Fuel for Off-Highway Diesel Engines” table in this Special Publication. Refer to Caterpillar S·O·S Services Fuel Analysis Section of this Chapter. Take corrective action if necessary. Corrective actions may include, but are not limited to, treating the fuel, cleaning of the fuel storage tank/system, and replacing the problematic fuel with fresh fuel.
- When fuels are stored for extended periods, follow all the fuel supplier and tank maintenance procedures. Circulate the fuel regularly through a filter to remove sediments. Test the fuel regularly for acid number, density, particle content, water and microbial growth. Observe trends of these properties to ensure no detrimental changes. The fuel is not recommended for use when its properties change negatively. For fuels containing biodiesel, the storage duration may be significantly reduced. Observe all the guidelines given in this section and in "Guidelines and potential impacts associated with the use of biodiesel and biodiesel blends" Table in the "Biodiesel" section of this Special publication.

NOTICE
In order to meet expected fuel system component life, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with common-rail fuel systems. Also, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with unit injected fuel systems. For all other Cat diesel engines (mostly older engines with pump, line and nozzle type fuel systems), the use of 4 micron(c) absolute or less secondary fuel filtration is strongly recommended. Note that all current Cat diesel engines are factory equipped with Cat Advanced Efficiency **and/or Ultra High Efficiency** filters 4 micron (c) absolute fuel filters.

Note: Thorough cleaning of fuel storage tanks is strongly recommended before converting to Ultra Low Sulfur Diesel (ULSD) (15 ppm or less sulfur) and/or biodiesel/biodiesel blends. Conversion to ULSD and/or biodiesel/biodiesel blends can loosen fuel system and fuel storage tank deposits. Bulk tank filtration unit and dispensing point filters, and onboard engine filters change intervals may need to be shortened for an extended period of time to allow for this cleaning effect.

PROOF

Note: Consult your local Cat dealer for additional information on Cat designed and produced filtration products.

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Fuel Information for Diesel Engines

SMCS Code: 1250; 1280

NOTICE
U. S. EPA regulations require the use of Ultra Low Sulfur Diesel fuel (ULSD) , ≤0.0015 percent (≤15 ppm (mg/kg)) sulfur, for nonroad and stationary Tier 4 EPA certified engines using fuel sensitive technologies such as SCR systems and particulate filters. Fuels other than ULSD can cause damage in those engines and should not be used.

Consult the U. S. EPA for fuel sulfur regulations and for the ULSD point of sales required dates for various nonroad applications.

European sulfur free fuel ≤0.0010 percent (≤10ppm (mg/kg) sulfur) fuel is required by regulation for use in engines certified to EU nonroad Stage IIIB and newer standards and that are equipped with exhaust aftertreatment systems.

Certain governments/localities and/or applications MAY require the use of ULSD fuel. Consult federal, state, and local authorities for guidance on fuel requirements for your area.

Typical aftertreatment systems include Diesel Particulate Filters (DPF), Diesel Oxidation Catalysts (DOC), Selective Catalytic Reduction (SCR) and/or Lean NOx Traps (LNT). Other systems may apply.

Low sulfur diesel (LSD) fuel (≤500 ppm (mg/kg) sulfur) is strongly recommended for use in engines that are pre-Tier 4 models while diesel fuel with >500 ppm sulfur is acceptable for use in areas of the world where allowed by law. Pre-Tier 4 engines that are equipped with a Diesel Oxidation Catalyst (DOC) require the use of LSD fuel or ULSD fuel.

ULSD fuel or sulfur-free diesel fuel are acceptable in all engines regardless of the engine U.S. EPA Tier or EU Stage requirements.

Use appropriate lubricating oils that are compatible with the engine certification and aftertreatment system and with the fuel sulfur levels. Refer to the “Diesel Fuel Sulfur Impacts” article of this “Fuels Specifications” section and to the “Lubricants Specifications” section of this Special Publication.

⚠ WARNING

Ultra Low Sulfur Diesel (ULSD) poses a greater static ignition hazard than earlier diesel formulations, with a higher sulfur content, which may result in a fire or explosion. Consult with your fuel or fuel system supplier for details on proper grounding and bonding practices.

Note: The removal of sulfur and other compounds in Ultra Low Sulfur Diesel (ULSD) fuel decreases the conductivity of ULSD and increases the ability of the fuel to store static charge. Refineries may have treated the fuel with a static dissipating additive. However, there are many factors that can reduce the effectiveness of the additive over time. Static charges can build up in ULSD fuel while the fuel is flowing through fuel delivery systems. Static electricity discharge when combustible vapors are present could result in a fire or explosion. Therefore, ensuring that the entire system used to refuel your machine (fuel supply tank, transfer pump, transfer hose, nozzle, and others) is properly grounded and bonded is important. Consult with your fuel or fuel system supplier to ensure that the delivery system is in compliance with fueling standards for proper grounding and bonding practices.

The two basic types of distillate diesel fuel are No. 2 diesel fuel and No. 1 diesel fuel. No. 2 diesel fuel is the most commonly available summer grade diesel fuel. No. 1 diesel fuel is a winter grade diesel fuel. During the winter months fuel suppliers will typically blend No. 1 and No. 2 diesel fuel in various percentages to meet the historical low ambient temperature cold-flow needs for a given area or region. No. 2 diesel fuel is a heavier diesel fuel than No. 1 diesel fuel. In cold weather, heavier fuels can cause problems with fuel filters, fuel lines, fuel tanks, and fuel storage. Heavier diesel fuels such as No. 2 diesel fuel can be used in diesel engines that operate in cold temperatures with an appropriate amount of a well proven pour point depressant additive. For more information on fuels which include blends of No. 1 and No. 2 diesel fuel, consult your fuel supplier.

When you use No. 2 diesel fuel or other heavier fuels, some of the fuel characteristics may interfere with successful cold-weather operation. Additional information about the characteristics of diesel fuel is available. This information contains a discussion on the modification to the characteristics of diesel fuel. There are several possible methods that can be used to compensate for the fuel qualities that may interfere with cold-weather operation. These methods include the use of starting aids, engine coolant heaters, fuel heaters, and de-icers. In addition, the manufacturer of the fuel can add cold flow improvers and/or blend No. 1 and No. 2 diesel in various percentages.

PROOF

Not all areas of the world classify diesel fuel using the No. 1 and No. 2 nomenclature described above. But, the basic principles of using additives and/or blending fuels of different densities to help compensate for the fuel qualities that may interfere with cold-weather operation are the same.

Starting Aids

The use of a starting aid is a conventional method of assistance for cold starts in low temperature conditions. Various starting aids are available for Caterpillar engines. Follow the recommendations that are provided by the manufacturer of the starting aid. Refer to the “Aftermarket Products and Warranty” article in the “ Warranty Information” section of this special publication.

Engine Coolant Heaters

These heaters heat the engine coolant. The heated coolant flows through the cylinder block. The flow of heated coolant keeps the engine warm. A warm engine is easier to start in cold weather. Most coolant heaters use electrical power. A source of electricity is necessary for this type of heater. Other heaters that burn fuel are available as a source of heat. These heaters may be used in place of the electrical heaters.

With either type of heater, starting aids and/or fuels with higher cetane numbers are less important because the engine is warm. Problems with fuel cloud point can cause the plugging of fuel filters. Problems with fuel cloud point cannot be corrected by engine coolant heaters. This is especially true for fuel filters that are cooled by air flow during operation.

Fuel Heaters

The fuel cloud point is related to problems with fuel filters. The fuel heater heats the fuel above the cloud point before the fuel enters the fuel filter. This prevents wax from blocking the filter. Fuel can flow through pumps and lines at temperatures below the cloud point. The cloud point is often above the pour point of a fuel. While the fuel can flow through these lines, the wax in the fuel can still plug the fuel filter.

In some engine installations, small modifications can prevent problems that are caused by the cloud point. One of the following changes can prevent problems in many conditions: a change in the location of fuel filters and/or supply lines and the addition of insulation. In extreme temperatures, heating of the fuel may be required to prevent the filters from plugging. There are several types of fuel heaters that are available. The heaters typically use either engine coolant or exhaust gas as a heat source. These systems may prevent filter waxing problems without the use of de-icers or cold flow improvers. These systems may be ineffective when the fuel contains a large amount of dirt or of water. Use of a fuel heater can help eliminate some cold-weather problems. A fuel heater should be installed so that the fuel is heated before flowing into the fuel filter.

Note: A fuel heater is not effective for cold-soaked starts unless the fuel heater can be powered from an external power source. External fuel lines may require the use of heaters that circulate the fuel.

Note: Only use properly sized fuel heaters that are controlled by thermostats or use fuel heaters that are self-regulated. Thermostatically controlled fuel heaters generally heat fuel to 15.5° C (60° F). Do not use fuel heaters in warm temperatures.

For distillate fuel configured engines, Caterpillar recommends a fuel viscosity as delivered to rotary fuel injection pumps of between 1.4 cSt and 4.5 cSt, and between 1.4 cSt and 20 cSt for all other fuel injection pumps.

Note: If a fuel with a low viscosity is used, cooling of the fuel may be required to maintain 1.4 cSt or greater viscosity at the fuel injection pump. Fuels with a high viscosity might require fuel heaters to lower the viscosity to either 4.5 cSt or less for rotary fuel injection pumps or 20 cSt viscosity or less for all other fuel injection pumps.

NOTICE
When you use fuel heaters, do not allow the fuel temperature to reach above 52°C (125°F). Never exceed 75° C (165° F) with straight distillate fuel. The high fuel temperatures affect the fuel viscosity. When the fuel viscosity falls below 1.4 cSt, pump damage may occur.

⚠ WARNING

Overheating the fuel or the fuel filter can result in personal injury and/or damage to the engine. Use extreme care and caution for heating of the fuel and/or the fuel filter.

PROOF

Select a fuel heater that is mechanically simple, yet adequate for the application. The fuel heater should also prevent overheating of the fuel. Disconnect the fuel heater or deactivate the fuel heater in warm weather. An unacceptable loss of fuel viscosity and engine power will occur if the fuel supply temperature is allowed to become too hot.

For additional information on fuel heaters, consult your Caterpillar dealer.

De-icers

De-icers lower the freezing point of the moisture in the fuel. De-icers are notgenerally needed when fuel heaters are used. If you experience trouble, consult your fuel supplier for recommendations of a compatible commercial de-icer.

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Characteristics of Diesel Fuel

SMCS Code: 1250; 1280

Viscosity

The viscosity of the fuel is significant because the fuel serves as a lubricant for fuel system components. Fuels need to have sufficient viscosity. The fuel must lubricate the fuel system in both extremely cold and in extremely hot temperatures.

Fuels of improper viscosity result in poor atomization and spray pattern when injected, which cause poor combustion and loss of performance. If the kinematic viscosity of the fuel is lower than 1.4 cSt as supplied to the fuel injection pump or to the unit injectors, excessive scuffing and seizure can occur. If the fuel viscosity is too high, the fuel may cause high fuel pump resistance, negatively impact the injector spray pattern, and may cause filter damage.

For distillate fuel configured engines, Caterpillar recommends a fuel viscosity as delivered to rotary fuel injection pumps of between 1.4 cSt and 4.5 cSt, and between 1.4 cSt and 20 cSt for all other fuel injection pumps.

If a fuel with a low viscosity is used, cooling of the fuel may be required to maintain 1.4 cSt or greater viscosity at the fuel injection pump. Fuels with a high viscosity might require heaters to lower the viscosity to either 4.5 cSt or less for rotary fuel injection pumps or 20 cSt or less for all other fuel injection pumps.

Cetane Number

The cetane number of the fuel effects the ability of the engine to start. Also, the cetane number effects the interval of time before the engine runs smoothly. Fuels of high cetane rating are easier to ignite. The starting temperature can be improved approximately 7 to 8°C (12 to 15°F) for every increase of ten in the cetane number. After the engine reaches the normal operating temperature, a change in the cetane from 40 to 50 will have a minimal effect on engine performance.

Most fuels that have a cetane number above 40 will permit acceptable engine starts in warmer outside temperatures. The engine will start satisfactorily with this fuel when the engine is kept warm. The engine can be kept warm by using either a heated enclosure or a properly sized coolant heater.

During average starting conditions, direct injection diesel engines require a minimum cetane number of 40. A higher cetane value may be required for operation in high altitudes or for cold-weather operation. The minimum fuel cetane number that is required for the precombustion chamber (PC) diesel engine is 35.

Modifying the Cetane Number

The cetane number of a fuel can be changed if the fuel is mixed with a fuel that has a different cetane number. Generally, the cetane number of the mixture will be in direct relation to the ratio of the fuels that were mixed. Your fuel supplier can provide the information about the cetane number of a particular fuel.

PROOF

Additives can also be used to improve the cetane number of a fuel. Additives are evaluated through testing in special test engines. However, the characteristics of fuels with natural cetane number can be different than those characteristics of a fuel addetized to reach the same cetane number. While both fuels may be rated as having the same cetane number, starting may be different.

Cloud Point

The cloud point of a fuel is different from the pour point. The cloud point is the temperature that allows some of the heavier components in the wax to solidify in the fuel. This wax is not a contaminant in the fuel. The wax is an important element of No. 2 diesel fuel. The wax has a high fuel energy content and the wax has a very high cetane value. Removal of the heavier wax lowers the cloud point of the fuel. Removal of the wax also increases the cost because less fuel can be made from the same amount of crude oil. Basically, a No. 1 diesel fuel is formulated by removing the wax from a No. 2 diesel fuel.

The cloud point of the fuel is important because the cloud point can limit the performance of the fuel filter. The wax can alter the fuel characteristics in cold weather. Solid wax can fill the fuel filters. The solidified wax will cause filter plugging. Plugged filters cannot remove contaminants from the fuel and hence cannot protect the fuel injection systems. Since fuel must flow through the filters, installing a fuel heater is the most practical way to prevent the problem. A fuel heater will keep the fuel above the cloud point as the fuel flows through the fuel system. The fuel heater will permit the wax to flow through the filters with the fuel.

Modifying the Cloud Point

You can lower the cloud point of a diesel fuel by mixing the diesel fuel with a different fuel that has a lower cloud point. No. 1 diesel fuel or kerosene may be used to lower the cloud point of a diesel fuel. The efficiency of this method is not good, because the ratio of the mixture does not have a direct relation to the improvement in cloud point. The amount of fuel with low cloud point that is required makes the process less preferable to use.

The fuel supplier must be consulted to provide the proper mix of fuels that offers the appropriate cloud point.

Another approach to modify the cloud point is to use cold flow improvement additives. The manufacturer of the fuel can add cold flow improvers to the fuel. Cold flow improvers modify the wax crystals in the fuels. The cold flow improvers do not change the cloud point of the fuel. However, the cold flow improvers keep the wax crystals small enough to pass through standard fuel filters. For mixing precautions, see the section “Pour Point”.

Generally, the most practical method that is used to prevent problems that are caused by fuel cloud point at low temperatures is the use of fuel heaters. In most applications, fuel heaters can be used at a lower cost than fuel mixtures.

The common standard methods that are used to test the cloud point of diesel fuels are:

- “ASTM D2500” Test Method for Cloud Point of Petroleum Products
- “ASTM D5771” Test Method for Cloud Point of Petroleum Products (Optical Detection Stepped Cooling Method)
- “ASTM D5772” Test Method for Cloud Point of Petroleum Products (Linear Cooling Rate Method)
- “ASTM D5773” Test Method for Cloud Point of Petroleum Products (Constant Cooling Rate Method)

Pour Point

The fuel's pour point is a temperature below the cloud point of the fuel. Fuel stops flowing below the pour point. The pour point is the temperature which limits movement of the fuel inside the pumps.

To measure the pour point, the fuel temperature is lowered below the cloud point in steps of 3°C (5°F) at a time. The temperature is lowered until the fuel does not flow. The pour point is the last temperature that is shown before the flow stops. At the pour point, the wax has solidified out of the fuel. This temperature makes the fuel more solid than liquid. The pour point of the fuel can be improved. This improvement does not require the removal of important elements. This process is the same process that is used to improve the cloud point of a fuel.

A fuel's pour point should be at least 6°C (10°F) below the lowest ambient temperature that is required for engine start-up and for engine operation. To operate the engine in extremely cold weather, No. 1 fuel or No. 1-D fuel may be necessary because of these fuels' lower pour points.

Modifying the Pour Point

You can lower the fuel's pour point by using additives. You can also lower the pour point of a diesel fuel by mixing the diesel fuel with a different fuel that has a lower pour point. No. 1 diesel fuel or kerosene may be used to lower the pour point of a diesel fuel. The amount of fuel with low pour point that is required makes the process less preferable to use.

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The following illustration contains a table that can be used to find the necessary mixture for two fuels with different pour points. This table is true only if the fuels do not have additives which change the pour point. This table may not apply to Ultra Low Sulfur diesel fuels and should be used only as a general guide. To use the table, you must know the exact pour point of each fuel. This specification can change from one purchase of fuel to the next purchase of fuel. This specification is normally available from personnel at the source of the fuel supply. When fuels that have a lower pour point are not available, this method cannot be used.

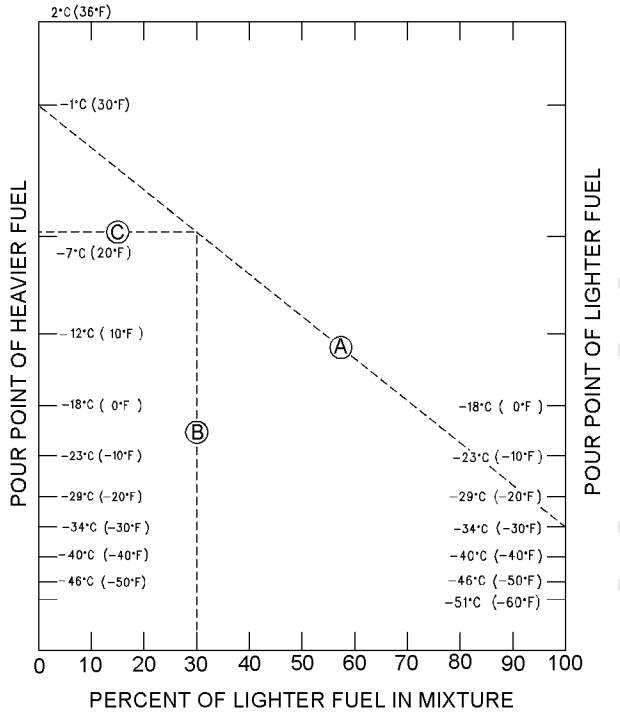


Illustration 2 g01180699

Pour point of fuel mixtures

To calculate the amount of lighter fuel that isrequiredto be blended with the heavier fuel, perform the following steps:

1. Obtain the specification for the cloud point or the pour point of both fuels from your fuel supplier.

2. Locate the cloud point or the pour point of the heavier fuel on the left side of the table. Mark the point on the table.

3. Locate the cloud point or the pour point of the lighter fuel on the right side of the table. Mark the point on the table.

4. Draw a line between the two points that were established. Label this line “A” .

5. Determine the lowest outside temperature for machine operation. Find this point on the left side of the table. Mark this point. Draw a horizontal line from this point. Stop the line at the intersection of line “A” . Label this new line “C” .

6. Line “C” and line “A” intersect. Mark this point. Draw a vertical line from this point. Stop the line at the bottom of the table. Label this line “B” . The point at the bottom of line “B” reveals the percentage of lighter fuel that is required to modify the cloud point or the pour point.

The above example shows that the blending will require a 30 percent mixture of lighter fuel.

Additives are a good method to use to lower the pour point of a fuel. These additives are known by the following names: pour point depressants, cold flow improvers, and wax modifiers. When the additives are used in the proper concentration, the fuel will flow through pumps, lines, and hoses.

Note: These additives must be thoroughly mixed into the fuel at temperatures that are above the cloud point. The fuel supplier should be contacted to blend the fuel with the additives. The blended fuel can be delivered to your fuel tanks.

The standard methods to measure the pour point of the fuels are:

- “ASTM D4539” Test Method for Filterability of Diesel Fuels by Low-Temperature Flow Test (LTFT)
- “ASTM D6371” Test Method for Cold Filter Plugging Point of Diesel and Heating Fuels

Lubricity and Low Sulfur Diesel (LSD) and Ultra Low Sulfur Diesel (ULSD) Fuel

The lubricity of the fluid describes the ability of the fluid to reduce the friction between surfaces that are under load. This ability reduces the damage that is caused by friction. Fuel injection systems rely on the lubricating properties of the fuel.

Note: The fuel lubricity is important. The lubricity of the fuel should be considered whenever you operate the equipment in temperature extremes, whether extremely hot or extremely cold. Also, you should consider the fuel lubricity whenever you use fuels that are lower in viscosity or that have been hydro-treated. There are many aftermarket additives that are available to treat fuel. If the lubricity of the fuel is an issue, consult your fuel supplier for proper recommendations regarding fuel additives. Also, refer to this Special Publication, “Distillate Diesel Fuel” article, “Aftermarket Fuel Additives” and “Cat Diesel Fuel Conditioner” topics.

The finished fuels as described by Caterpillar Diesel Fuel Specification, “ASTM D975” or “EN 690” are at the recommended lubricity levels. To determine the lubricity of the fuel, use the “ASTM D6079 High Frequency Reciprocating Rig (HFRR)” test. The maximum allowable wear scar is 0.52 mm (0.0205 inch) at 60° C (140° F). If the lubricity of a fuel does not meet the minimum requirements, consult your fuel supplier. Do not treat the fuel without consulting the fuel supplier. Some additives are not compatible. These additives can cause problems in the fuel system.

The process that is most commonly used to remove sulfur from fuel is called hydro-treatment. This process is also the most economical process. Each source of crude oil contains different amounts of sulfur. Crude oils typically require hydro-treatment to obtain the 0.0015 percent maximum sulfur limit. Crude oils with high sulfur require a more severe treatment.

The hydro-treatment removes the sulfur and other components from the fuel. The treatment removes nitrogen compounds, polar materials, bicyclic aromatics, polycyclic aromatics, and oxygen compounds. While the removal of sulfur has shown no detrimental effects to the engine, the removal of other compounds have lowered the lubricity of the fuel. As a result of the lowered lubricity, the fuel is less tolerant of contamination by water and dirt. The lower fuel lubricity can be seen as abrasive wear of fuel system components. Fuels that have a low lubricity may not provide adequate lubrication to plungers, to barrels, and to injectors. This problem may be compounded in areas that require winter blends of fuel. The lighter winter fuel blend has the following characteristics: lower viscosity, lower cloud point and lower pour point.

The finished fuels that are per the recommended specifications should have the correct lubricity. However, if required, the lubricity of the fuel may be enhanced with additives. Many fuel suppliers treat the fuel with these additives. Do not use a fuel lubricity additive before you consult the fuel supplier. Some aftermarket additives may not be compatible with the additives that are already in the fuel, and some may damage emission control systems. Some additive packages that are supplied by the aftermarket manufacturer may not be compatible with the seals that are used in fuel systems of some diesel engines. Other additive packages that are supplied by aftermarket manufacturers cannot provide proper performance in high temperature conditions. These additives may leave deposits because of the high temperatures that exist in the fuel systems of diesel engines.

Maximum life of the fuel system can be achieved by performing the following tasks: using a preferred distillate diesel fuel (refer to the “Fuel Recommendations” article in this Special Publication), using a reliable fuel supplier and performing proper maintenance of the fuel system. Caterpillar Advanced Efficiency fuel filters are required for diesel engines that run on diesel fuel to provide maximum life to the fuel system.

Note: Lighter fuels are frequently used in arctic temperatures. Lighter fuels may include the following fuels: Jet A, Jet A-1, JP-8, JP-5, and kerosene. The specifications that apply to these fuels do not include a minimum lubricity requirement. Do not assume that a fuel meets the minimum Caterpillar specification. Contact the fuel supplier for proper recommendations on fuel lubricity additives.

Note: The sulfur levels for Jet A, Jet A-1, JP-8, JP-5, and kerosene fuels typically far exceed 15 ppm, the U.S. ULSD fuel, and the sulfur levels for these fuels typically far exceed 50 ppm, the EU low sulfur fuel.

Note: For best results, your fuel supplier should treat the fuel when additives are required.

Refer to this Special Publication, “Distillate Diesel Fuel” article, “Aftermarket Fuel Additives”, “Cat Diesel Fuel Conditioner”, and “Alternative Fuels - Arctic Applications” topics for guidance.

Fuel volatility

Fuel volatility is measured and controlled by the fuel distillation curve. The optimal fuel volatility required for various engines depends on the engine application, design, loads, speeds, ambient temperatures, and other factors. Low volatility fuels may have a higher energy content (heating value). On the other hand, fuels of high initial volatility may improve the startability and warm up process and reduce smoke. High-performance fuels have the right balance of volatility.

The fuel distillation curve describes the amount of fuel that evaporates at various temperatures. Of these temperatures, the heavy end is characterized by the T90, the temperature where 90 percent of the fuel evaporates. If the T90 exceeds the maximum limits given in the "Cat Specification for Distillate Fuel for Nonroad Diesel Engines" Table, in the "Distillate Diesel Fuel" section, the fuel may increase smoke, deposits, soot, and particulate matter emissions. The lower end or low distillation temperatures are not specified in the "Cat Specification for Distillate Fuel for Nonroad Diesel Engines" Table, nor in “ASTM D975” or similar specifications. However, very low distillation temperatures may cause the fuel to become volatile at low temperatures and may cause cavitation of fuel pumps or fuel system components.

Diesel Fuel Sulfur

Sulfur is a natural component of diesel fuels. High sulfur in the fuel can be reduced through refining technologies.

Sulfur levels in the fuel affect the durability of engine components and also affect engine exhaust emissions. Modern Cat diesel engines are designed to meet mandated gaseous emissions requirements. To meet these emissions requirements, the engines are tested and developed with specific sulfur levels in the diesel fuel.

The maximum allowable fuel sulfur level is controlled by various emissions laws, regulations, and mandates. Consult federal, state, and local authorities for guidance on fuel requirements for your area.

The list below provides a quick reference for acceptable sulfur levels for diesel fuel that will be used in Cat machine diesel engines but the controlling documents are the engine Operation and Maintenance Manuals, the specific aftertreatment device documentation, and the applicable emissions laws, regulations, and mandates.

- U.S. EPA regulations require the use of Ultra Low Sulfur Diesel fuel (ULSD) , ≤ 0.0015 percent (≤ 15 ppm (mg/kg)) sulfur, for nonroad and stationary Tier 4 EPA certified engines using fuel sensitive technologies such as SCR systems and particulate filters. Fuels other than ULSD can cause damage in those engines and should not be used. Consult the U.S. EPA for fuel sulfur regulations and for the ULSD point of sales required dates for various nonroad applications.
- European sulfur free fuel, 0.0010 percent (= 10 mg/kg) sulfur, fuel is required by regulation for use in engines certified to EU nonroad Stage IIIB and newer standards and that are equipped with exhaust aftertreatment systems.
- Certain governments/localities and/or applications MAY require the use of ULSD fuel. Consult federal, state, and local authorities for guidance on fuel requirements for your area.
- The maximum allowable fuel sulfur level for most pre-Tier 4 engines that are equipped with DOC (Diesel Oxidation Catalyst) is 0.05 percent (500 ppm (mg/kg)). Some DOC equipped engines require the use of fuel with a maximum of 0.005% (50 ppm (mg/kg)) fuel sulfur. Refer to the engine/machine Operation and Maintenance Manual and refer to the aftertreatment device-specific documentation for guidance.
- For machine diesel engines that are retrofitted with an aftertreatment device, refer to the aftertreatment device-specific documentation.

Typical aftertreatment systems include Diesel Particulate Filters (DPF), Diesel Oxidation Catalysts (DOC), Selective Catalytic Reduction (SCR) and/or Lean NOx Traps (LNT). Other systems may apply.

In addition to the emission regulations, factors that affect maximum allowed and/or acceptable fuel sulfur level include:

- Engine model/design
- Engine application
- Overall fuel quality
- Using recommended fluids, including but not limited to engine oil quality
- Exhaust aftertreatment device type
- Environmental factors and other site-specific operating conditions
- Fuel costs versus risk of shortened engine/engine component life
- Fuel costs versus shortened oil drain intervals
- Maintenance intervals and other maintenance practices

Ultra-Low Sulfur Diesel (ULSD)

The United States (U.S.) Environmental Protection Agency (EPA) defines Ultra-Low Sulfur Diesel (ULSD - S15) as a U.S. diesel fuel with a sulfur content not to exceed 15 parts per million (ppm(mg/kg)) or 0.0015 percent by weight.

ULSD was introduced for the U.S. on-highway diesel engine market in October 2006. ULSD is available sinceDecember 2010for nonroad diesel engines and machines. Refer to the U.S. EPA for the required ULSD point of sales dates for various nonroad applications.

PROOF

Engines certified to nonroad Tier 4 standards (Stage IV in Europe) and are equipped with fuel sulfur sensitive exhaust aftertreatment systems are designed to run on ULSD only. Use of LSD or fuels higher than 15 ppm (mg/kg) sulfur in these engines will reduce engine efficiency and engine durability and will damage emissions control systems and/or shorten the service interval. Failures that result form the use of fuels are not Cat factory defects. Therefore the cost of repairs would not be covered by a Cat warranty.

ULSD fuel can be used in any engine designed to run on diesel fuel. Cat does not require the use of ULSD in nonroad and machine applications that are not Tier 4/Stage IIIB/Stage IV certified engines and are not equipped with aftertreatment devices. For Tier 4/ Stage IIIB/Stage IV certified engines, always follow operating instructions and fuel tank inlet labels, if available, to insure the correct fuels are used.

Note: The removal of sulfur and other compounds in Ultra Low Sulfur Diesel (ULSD) fuel decreases the conductivity of ULSD and increases the ability of the fuel to store static charge. Refineries may have treated the fuel with a static dissipating additive. However, there are many factors that can reduce the effectiveness of the additive over time. Static charges can build up in ULSD fuel while the fuel is flowing through fuel delivery systems. Static electricity discharge when combustible vapors are present could result in a fire or explosion. Therefore, ensuring that the entire system used to refuel your machine (fuel supply tank, transfer pump, transfer hose, nozzle, and others) is properly grounded and bonded is important. Consult with your fuel or fuel system supplier to ensure that the delivery system is in compliance with fueling standards for proper grounding and bonding practices.

The standard methods for testing conductivity of diesel fuel are:

- “ASTM D2624” Test Methods for Electrical Conductivity of Aviation and Distillate Fuels
- “ASTM D4308” Test Method for Electrical Conductivity of Liquid Hydrocarbons by Precision Meter

Sulfur-free Diesel Fuel

In Europe, ultra low sulfur diesel fuel will have a maximum of 0.0010 percent (10 ppm(mg/kg)) sulfur and is typically referred to as “sulfur-free”. This sulfur level is defined in “European Standard EN 590:2004”.

Low Sulfur Diesel (LSD)

Low Sulfur Diesel (LSD - S500) is defined by the U.S. EPA as a U.S. diesel fuel with sulfur content not to exceed 500 ppm or 0.05 percent by weight.

Note: Both ULSD and LSD must meet the fuel requirements outlined in the most current revision level of “ASTM D975”.

Diesel Fuel Sulfur Impacts

Sulfur in the fuel results in the formation of sulfur dioxide (SO2) and sulfur trioxide (SO3) gases during the combustion process. When combined with water in the exhaust gas SO2 and SO3 can form acids. The acids can impact engine components and engine lubricants.

Sulfur in the exhaust gas can interfere with the operation of aftertreatment devices causing loss of passive regeneration performance, reduced gaseous emission conversion efficiency, and increased particulate matter emissions.

Typical aftertreatment systems include Diesel Particulate Filters (DPF), Diesel Oxidation Catalysts (DOC), Selective Catalytic Reduction (SCR) and/or Lean NOx Traps (LNT). Other systems may apply.

Use of fuels with higher than recommended and/or maximum allowed fuel sulfur levels can and/or will:

- Increase wear of engine components
- Increase corrosion of engine components
- Increase deposits
- Increase soot formation
- Shorten the time period between oil drain intervals (cause the need for more frequent oil drain intervals)
- Shorten the time interval between aftertreatment device service intervals (cause the need for more frequent service intervals)
- Negatively impact the performance and life of aftertreatment devices (cause loss of performance)
- Reduce regeneration intervals of aftertreatment devices
- Lower fuel economy
- Increase overall operating costs

Depending on operating conditions, and depending on maintenance practices, the potential issues stated above may and/or will take place with fuel sulfur levels that are at or below the recommended fuel sulfur levels, and/or that are at or below the maximum allowable fuel sulfur levels.

Fuel sulfur levels above 0.1% (1000 ppm (mg/kg)) may significantly shorten the oil change interval.

PROOF

When other factors do no preclude, and understanding that there may be trade-offs such as shortened oil drain intervals, certain commercial, and machine diesel engines that are covered by this Special Publication MAY be able to operate satisfactorily on fuels with up to 1 percent (10, 000 ppm(mg/kg)) sulfur if the following conditions are met:

- All emissions laws, regulations, and mandates are followed
- The engine/engines are not equipped with aftertreatment device/devices
- All appropriate guidelines and maintenance practices as stated in the engine Operation and Maintenance Manual are followed
- All appropriate guidelines and maintenance practices as stated in this Special Publication are followed
- Operating in otherwise low to moderate severity applications
- Your Cat dealer is consulted and approves
- You refer to this Special Publication, and you refer to your specific Cat commercial engine and/or refer to your specific Cat machine Operation and Maintenance Manual for more guidance and exceptions

Oil Drain Intervals

Note: DO NOT USE ONLY THIS SPECIAL PUBLICATION AS A BASIS FOR DETERMINING OIL DRAIN INTERVALS.

Fuel sulfur level impacts the oil drain interval. For detailed information, refer to the “S·O·S Services Oil Analysis” section in the “Lubricants Specification” article in this Special Publication.

- Cat S·O·S Services oil analysis is recommended.
- Cat S·O·S Services oil analysis is verystronglyrecommended to determine oil drain intervals when using fuel with sulfur levels between 0.05% (500 ppm) and 0.5% (5000 ppm).
- Cat S·O·S Services oil analysis is required to determine oil drain intervals when using fuel with sulfur levels above 0.5% (5000 ppm).
- Consult your Cat dealer for guidance when fuel sulfur levels are above 0.1% (1000 ppm).

Moisture Content

Problems with fuel filters can occur at any time. The cause of the problem can be water in the fuel or moisture in the fuel. At low temperatures, moisture causes special problems. There are three types of moisture in fuel: dissolved moisture (moisture in solution), free and dispersed moisture in the fuel and free and settled at the bottom of the tank.

Most diesel fuels have some dissolved moisture. Just as the moisture in air, the fuel can only contain a specific maximum amount of moisture at any one temperature. The amount of moisture decreases as the temperature is lowered. For example, a fuel could contain 100 ppm(100 mg/kg or 0.010 percent) of water in solution at 18°C (65°F). This same fuel can possibly hold only 30 ppm(30 ppm or 0.003 percent) at 4°C (40°F).

After the fuel has absorbed the maximum amount of water, the additional water will be free and dispersed. Free and dispersed moisture is fine droplets of water that is suspended in the fuel. Since the water is heavier than the fuel, the water will slowly become free and settled at the bottom of the tank. In the above example, when the fuel temperature was lowered from 18°C (65°F) to 4°C (40°F), 70 ppm (mg/kg) of water became free and dispersed in the fuel.

The small drops of water cause a cloudy appearance in the fuel. If the change in temperature is slow, the small drops of water can settle to the bottom of the tank. When the fuel temperature is lowered rapidly to freezing temperature, the moisture that comes out-of-solution changes to very fine particles of ice instead of small drops of water.

The particles of ice are lighter than the fuel, and the particles of ice will not settle to the bottom of the tank. When this type of moisture is mixed in the fuel, this moisture will fill the fuel filters. The ice crystals will plug the fuel filters in the same way as wax plugs the fuel filters.

If a filter is plugged and fuel flow is stopped, perform the following procedure to determine the cause:

1. Remove the fuel filters.

2. Cut the fuel filters open.

3. Inspect the fuel filter before the filter warms. This inspection will show that the filter is filled with particles of either ice or wax.

PROOF

The moisture which is free and settled at the bottom of the tank can become mixed with the fuel. The force of any pumping action will mix the moisture with the fuel whenever fuel is transferred. This moisture then becomes free and dispersed water. This moisture can cause ice in the filters. This moisture can cause other problems with filters at any temperature. Generally, the same force that mixes the water into the fuel will also mix dirt and rust from the bottom of the tank with the water. The result is a dirty mixture of fuel and water which can also fill the filters and stop fuel flow.

Specific Gravity / API Gravity

The specific gravity of diesel fuel is the weight of a fixed volume of fuel in comparison to the weight of the same volume of water at the same temperature. A higher specific gravity correlates into a heavier fuel. Heavier fuels have more energy or power per volume for the engine to use.

Note: The settings for the fuel mixture should not be adjusted to compensate for a loss of power with fuels that are lighter. The life of fuel system components can be decreased with fuels that are very light because lubrication will be less effective as a result of the lower viscosity. This issue is compounded if the fuel does not have sufficient lubricity. Refer to the “Lubricity and Low Sulfur Fuel Diesel (LSD) and Ultra Low Sulfur Diesel (ULSD) Fuel” topic in this Special Publication, “Characteristics of Diesel Fuel” article.

The API gravity of a fuel is also a measure of the density of the fuel or the relationship of the weight to the volume. The scale for API gravity is inverse to the scale for specific gravity. The API gravity will become higher as the fuel becomes lighter.

Lighter fuels will not produce the rated power. Lighter fuels may also be a blend of ethanol or methanol with diesel fuel. Blending alcohol or gasoline with diesel fuel will create an explosive atmosphere in the fuel tank. In addition, water condensation in the tank can cause the alcohol to separate in the tank.

⚠ WARNING

Mixing alcohol or gasoline with diesel fuel can produce an explosive mixture in the engine crankcase or fuel tank.

Personal injury and damage to the engine may result. Caterpillar recommends against this practice.

NOTICE

Mixing alcohol or gasoline with diesel fuel may cause damage to the engine. Caterpillar recommends against this practice. Water condensation in the fuel tank can cause the alcohol to separate which could cause damage to the engine.

Heavier fuels tend to create more deposits from combustion. Deposits from combustion can cause abnormal cylinder liner and ring wear. This problem is most noticeable in smaller diesel engines that operate at higher speeds.

Gums and Resins

The gums and resins that occur in diesel fuel are the result of dissolved oxidation products in the fuel that do not evaporate easily. The products that are dissolved in the fuel also do not burn cleanly. Excessive gum in the fuel will coat the inside of the fuel lines, pumps, and injectors. Excessive gum will also interfere with the close tolerances of the moving parts of the fuel systems. Gum and resin in the fuel will also cause the filter to plug rapidly. Oxidation of the fuel will occur and the formation of more gums and resins will occur during fuel storage. The storage time for fuel needs to be minimized to help reduce the formation of gums and resins.

Note: Even when all fuel storage maintenance practices that are relevant to your application are followed, Caterpillar recommends a maximum of 1 year from production for distillate diesel fuel storage, and a maximum of 6 months from production for biodiesel and blended biodiesel storage. Storage life for biodiesel and biodiesel blends that are greater than B20 may be much shorter than 6 months.

The Thermal Stability and Oxidation Stability of Fuel

Diesel fuels can deteriorate rapidly for various reasons. When the fuel is stressed and stored for long intervals, degradation and oxidation can occur. Degradation and oxidation are complex chemical changes, which may include the formation of peroxides. These changes lead to deposits or sediment from certain hydrocarbons and traces of naturally occurring nitrogen and sulfur containing compounds in the fuel. Fuel composition and environmental factors influence the process.

Diesel fuel is being used as a coolant for high-pressure fuel injection systems with high temperature fuel wetted walls. This process can stress the fuel in the fuel system. The thermal stress and an increase in recirculation fuel temperature is often responsible for fuel degradation and the formation of gums, resins, sediment, and deposits, which can cause fuel flow restriction through fuel filters and fuel injection systems.

When a fuel is left in a machine or engine fuel tank for a long time, the fuel is exposed to oxygen. This exposure leads to complex chemical reactions and degradation of the fuel. As a result, sludge and deposits are formed, which lead to poor performance, filter plugging, restriction of fuel lines, and deposits in the injector.

PROOF

Biodiesel and blends of biodiesel have poor thermal stability and oxidation stability compared to petroleum distillate diesel fuels. The use of these biodiesels and blends of biodiesel can accelerate the problems that are addressed in this Special Publication. Using biodiesel blends above the maximum level approved for the engine is not recommended.

Thermal and oxidative degradation of diesel fuel can result in a darkening of fuel color. Fuel color is not necessarily an indication of excessive degradation that will lead to the problems outlined in this Special Publication. But darkened fuel color can be an indicator of degradation leading to concerns about the stability of darkened fuel. Thermal oxidation and oxidative stability tests should be run to confirm actual fuel degradation.

Testing the fuels for thermal and oxidative stability as described in the"Cat Specification for Distillate Fuel for Nonroad Diesel Engines" Table, in the "Distillate Diesel Fuel" section, ensures that the fuel meets the minimum requirements for stability. Fuels that pass these tests offer the desired performance and reduce the deposit formation.

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Fuel Recommendations

SMCS Code: 1250; 1280

NOTICE
These recommendations are subject to change without notice. Consult your local Cat dealer for the most up to date recommendations.

Diesel engines may burn a wide variety of fuels. These fuels are divided into two general groups. The two groups are called the preferred fuels and the permissible fuels.

The preferred fuels provide maximum engine service life and performance. The preferred fuels are distillate fuels. These fuels are commonly called diesel fuel, furnace oil, gas oil, or kerosene. These fuels must meet the “Cat Specification for Distillate Diesel Fuel for Off-Highway Diesel Engines” found in this Special Publication, “Distillate Diesel Fuel” article.

The permissible fuels are some crude oils, some blends of crude oil with distillate fuel, and some marine diesel fuel. **These fuels are not suitable for use in all engine applications.** The acceptability of these fuels for use is determined on a case by case basis. A complete fuel analysis is required. Consult your Cat dealer for further information. Biodiesel fuel is permissible for use in Cat engines. Follow all the recommendations and guidelines given in this Special Publication, “Biodiesel” article.

Note: Except for some biodiesel, permissible fuels are not acceptable for use in on-highway applications.

NOTICE
Use of permissible fuels can result in higher maintenance costs and reduced engine service life.

Note: Use of fuels that do not meet at least the minimum performance recommendations and/or requirements may lead to lower compartment performance and/or compartment failure. Problems/failures that are caused by using fuels that do not meet the minimum recommended and/or required performance level are not Cat factory defects and therefore are NOT covered by the Cat warranty. The fuel supplier and customer are responsible.

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Distillate Diesel Fuel

SMCS Code: 1280

Note: For on-highway diesel engine fluids requirements, refer to specific engine Operation and Maintenance Manuals, and also refer to the most current revision level of Special Publication, SEBU6385, “Caterpillar On-Highway Diesel Engine Fluids Recommendations”. Also consult your Cat dealer.

Caterpillar is not in the position to continuously evaluate and monitor all the many worldwide distillate diesel fuel specifications and the on-going revisions that are published by governments and technological societies.

The “Caterpillar Specification for Distillate Fuel for Off-Highway Diesel Engines” provides a known, reliable baseline to judge the expected performance of distillate diesel fuels that are derived from conventional sources (crude oil, shale oil, oil sands, etc.) when used in Cat diesel engines.

Using the Cat distillate diesel fuel specification as the baseline, it is much easier to determine any potential economic and/or performance trade-offs, and overall acceptability when using fuels of varying characteristics and quality levels.

- When required, have the diesel fuel that either is being used or is planned to be used, tested per the Cat distillate diesel fuel specification.
- Use the Cat distillate diesel fuel specification as a fuel quality baseline for comparison of distillate diesel fuel analysis results, and/or a baseline for comparison of other distillate diesel fuel specifications.

PROOF